

Chapter 5: Operators on Inner Product Spaces

episode 10.1

def 5.1 | given a linear map $\alpha: V \rightarrow W$ of inner product spaces, the **adjoint** of α is the function $\alpha^*: W \rightarrow V$ such that

$$\langle \alpha(v), w \rangle_W = \langle v, \alpha^*(w) \rangle_V$$

for all $v \in V, w \in W$

lemma 5.3 | Let $\alpha: V \rightarrow W$ be a linear map of inner product spaces. Then, the adjoint $\alpha^*: W \rightarrow V$ is a linear map

lemma 5.4 | Let $\alpha, \beta: V \rightarrow W, \gamma: W \rightarrow U$ and $\lambda \in \mathbb{F}$. Then

(a) $(\alpha + \lambda\beta)^* = \alpha^* + \overline{\lambda}\beta^*$

(b) $(\alpha^*)^* = \alpha$

(c) for the identity map $\text{id}^* = \text{id}$

(d) $(\gamma \circ \alpha)^* = \alpha^* \circ \gamma^*$

(e) $\text{Ker}(\alpha^*) = \text{Im}(\alpha)^\perp$

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(g) $\text{Ker}(\alpha) = \text{Im}(\alpha^*)^\perp$

(h) $\text{Im}(\alpha) = \text{Ker}(\alpha^*)^\perp$

def 5.5 | an operator $\alpha: V \rightarrow V$ is **self-adjoint** if $\alpha = \alpha^*$. In other words, for all $v, w \in V$,

$$\langle \alpha(v), w \rangle = \langle v, \alpha(w) \rangle$$

ex 5.6 | Taking $V = \mathbb{R}^2$ w/ the Euclidean inner product, the map

$$\alpha \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x + 3y \\ 3x + 7y \end{pmatrix} \text{ is self adjoint but}$$

$$\beta \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x + 9y \\ 3x + 7y \end{pmatrix} \text{ is not}$$

lemma 5.7 | if α is a self-adjoint operator on V and for every $v \in V$, $\langle \alpha(v), v \rangle = 0$, then α is the zero map

def 5.8 | an operator $\alpha: V \rightarrow V$ is called **normal** if $\alpha^* \circ \alpha = \alpha \circ \alpha^*$

lemma 5.9 | an operator α is normal iff for every $v \in V$,

$$\|\alpha(v)\| = \|\alpha^*(v)\|$$

episode 10.2

lemma 5.10 | if α is a normal operator, then so is $\alpha - \lambda \text{id}$ for any $\lambda \in \mathbb{F}$

lemma 5.11 | let α be a normal operator on V , and $v \in V$ an eigenvector of α with eigenvalue λ . Then v is also an eigenvector of α^* w/ eigenvalue $\overline{\lambda}$

lemma 5.12 | Suppose α is a normal operator. If u, v are eigenvectors of α corresponding to distinct eigenvalues $\lambda \neq \mu$, then $\langle u, v \rangle = 0$

lemma 5.13 | to get the matrix of α^* , one transposes and then takes the complex conjugate of α

thm 5.14 | **the complex spectral theorem** - let α be an operator on a finite dimensional complex inner product space V . then, α is normal iff V has a basis consisting of orthogonal eigenvectors of V

lemma 5.15 | let α be a self-adjoint operator on V , and let b and c be real numbers such that $b^2 - 4c > 0$. Then, the linear map $\alpha^2 + b\alpha + c\text{id}$ is invertible

lemma 5.16 | let V be a real nonzero vector space and α a self-adjoint operator on V . Then, α has an eigenvalue

lemma 5.17 | let α be a self-adjoint operator on V and let U be an invariant subspace of α . Then:

- (a) $U^\perp$ is invariant under α
- (b) α restricts to a self adjoint operator on U

thm 5.18 | **Real Spectral Theorem**

let α be an operator on a finite-dimensional real vector space V . Then, V has an orthonormal basis consisting of eigenvectors of α iff α is self-adjoint